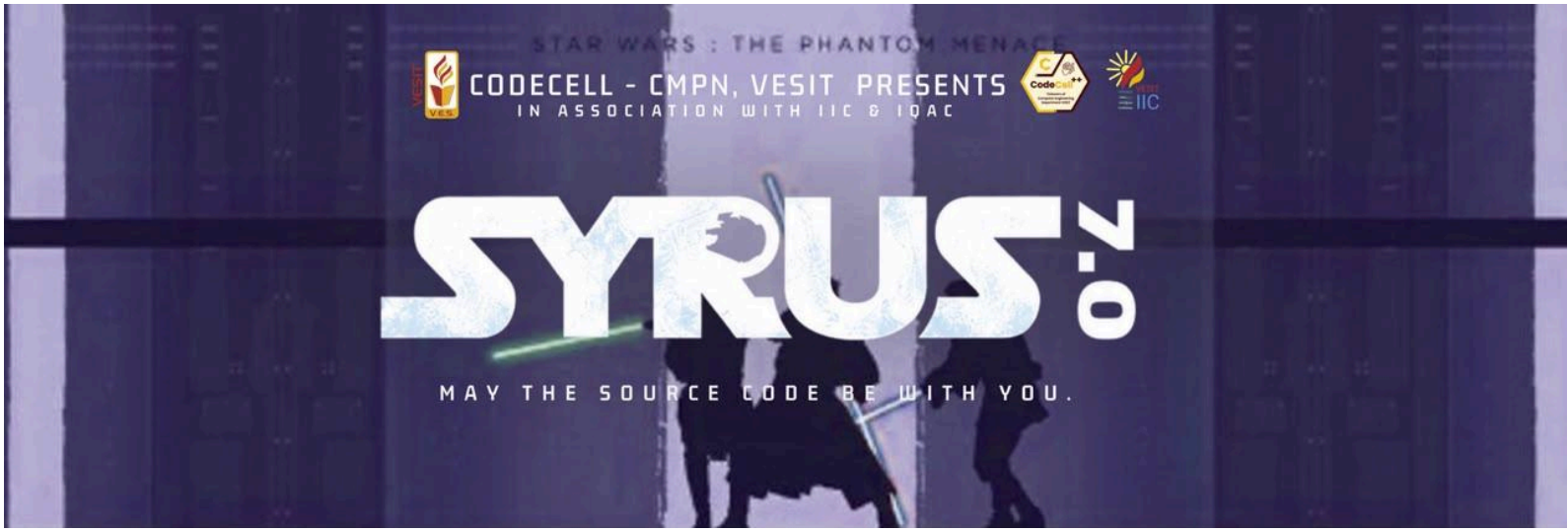


Title: AI Market and Risk Intelligence for Agri Supply Chains - FASALYTICS

Domain: Sustainability / Agriculture Technology

Problem Statement:

Farmers often have access to mandi prices but lack actionable intelligence on **where, when and at what price to sell**. The proposed system uses AI to forecast mandi prices, compare markets, calculate expected net returns and assess market risks.

Abstract:

FASALYTICS transforms agricultural market data into personalized selling decisions for farmers. The system analyzes mandi prices, arrival volumes, seasonality, market trends, transportation costs, storage/spoilage costs, weather signals and volatility to forecast prices for the next **1-7 days**.

Rather than recommending the highest-priced mandi, the system calculates **expected net return** after logistics and spoilage costs and recommends the best risk-adjusted selling option. It compares multiple mandis and can recommend **when to sell, where to sell and how much produce to send to each market**.

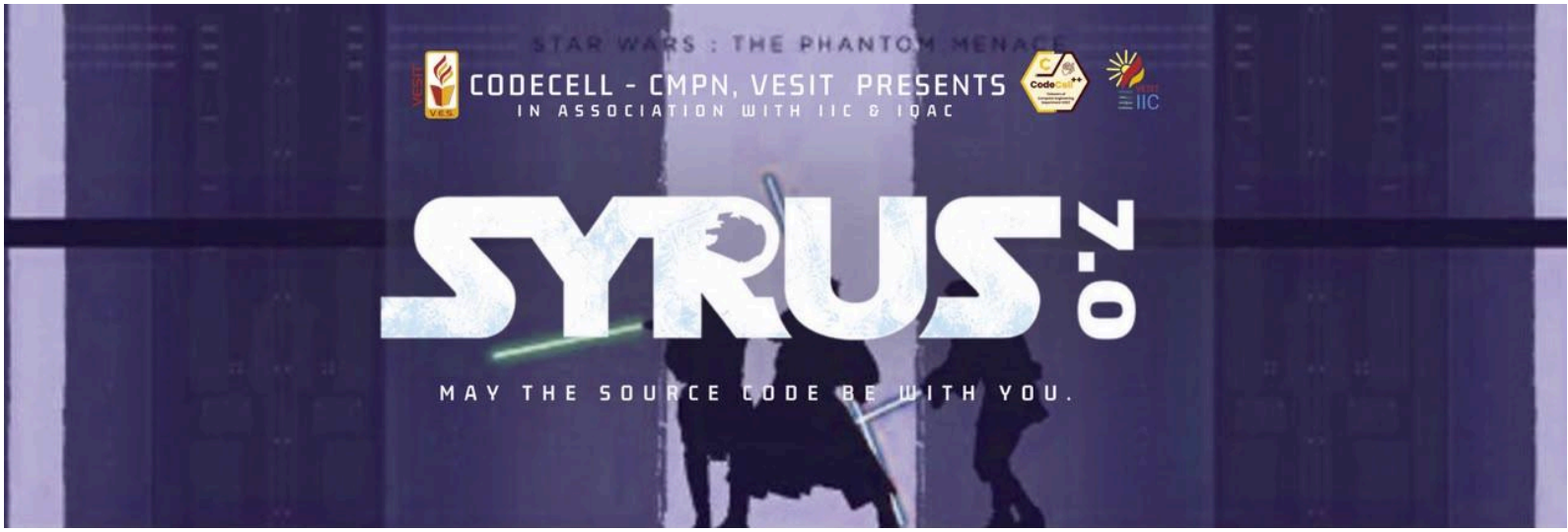
A key innovation is **Crowd-Aware Mandi Recommendation**, where the system considers expected farmer arrivals and avoids directing excessive farmers toward the same market. A **Fair Price Card** also provides an estimated fair-price range to help farmers negotiate with traders.

To improve accessibility, the same AI intelligence can be accessed through a **Web Dashboard, WhatsApp, or ordinary SMS**, supporting Marathi, Hindi, English and code-mixed conversations.

Expected Features

- Market Data Collection & Quality**:** Historical and recent prices and arrivals with cleaning, standardization, source and timestamp tracking.
- Short-Term Price Forecasting: 1-7 day price forecasting
- Smart Mandi Recommendations
- Net-return optimization using logistics and spoilage costs
- LLM Assistant & Scenario Analysis**:** Tool-connected AI that recalculates forecasts, costs, risks and recommendations for what-if scenarios.
- LLM Assistant and Scenario Analysis:Conversational AI with tool-connected forecasting, recommendation engines and what-if/scenario analysis
- Interactive Dashboard
- Weather and regional-language integration





Key Innovations

- **Progressive Web App:** Mobile-first, responsive, and installable application accessible through a web browser
- **Two-Way SMS AI:** Farmers can access recommendations through ordinary SMS without smartphones or internet.
- **Fair Price Card:** Provides an estimated fair-price range to support trader negotiations.
- **Crowd-Aware Mandi Recommendation:** Considers expected arrivals and avoids directing too many farmers to one market.
- **Smart Harvest Allocation:** Suggests splitting produce across multiple mandis for better risk-adjusted returns.
- **Proactive Risk Alerts:** Warns farmers about potential price crashes, arrival spikes and volatility.
- **Weather-to-Market Intelligence:** Links weather → transport → arrivals → price/risk.
- **Explainable AI:** Shows why a particular mandi and selling time are recommended.
- **Farmer Feedback Loop:** Compares predicted prices with actual selling prices to improve future decisions.
- **Multilingual & Code-Mixed AI:** Supports Marathi, Hindi, English and mixed-language conversations.

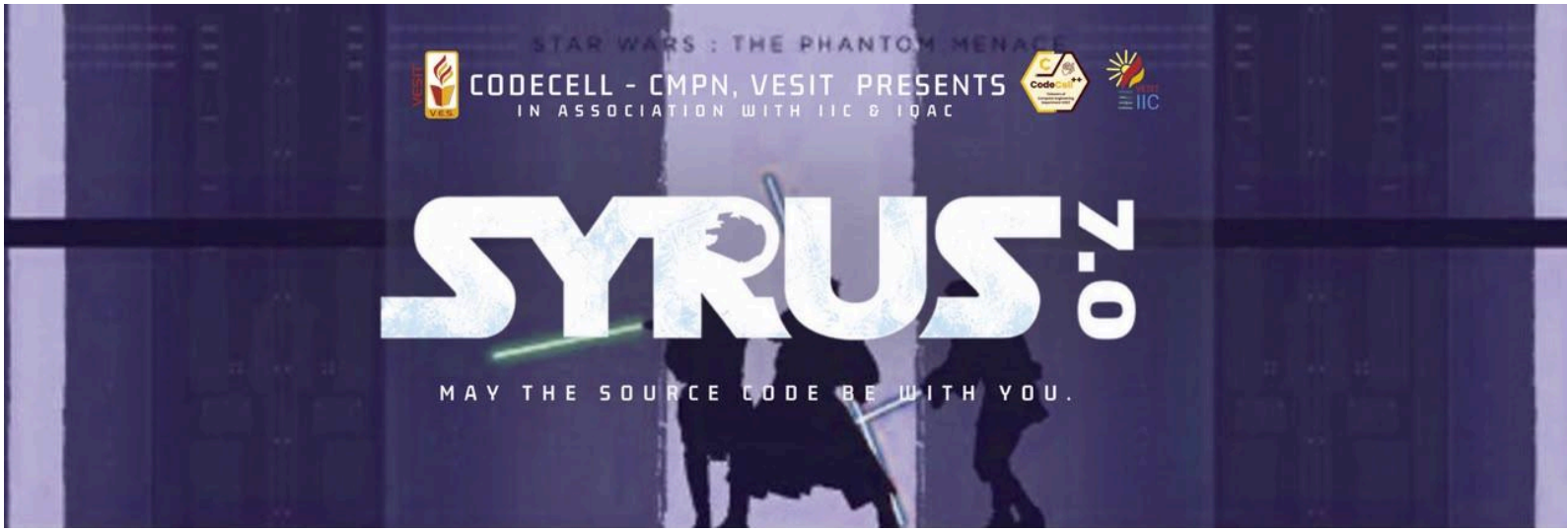
Tech Stack:

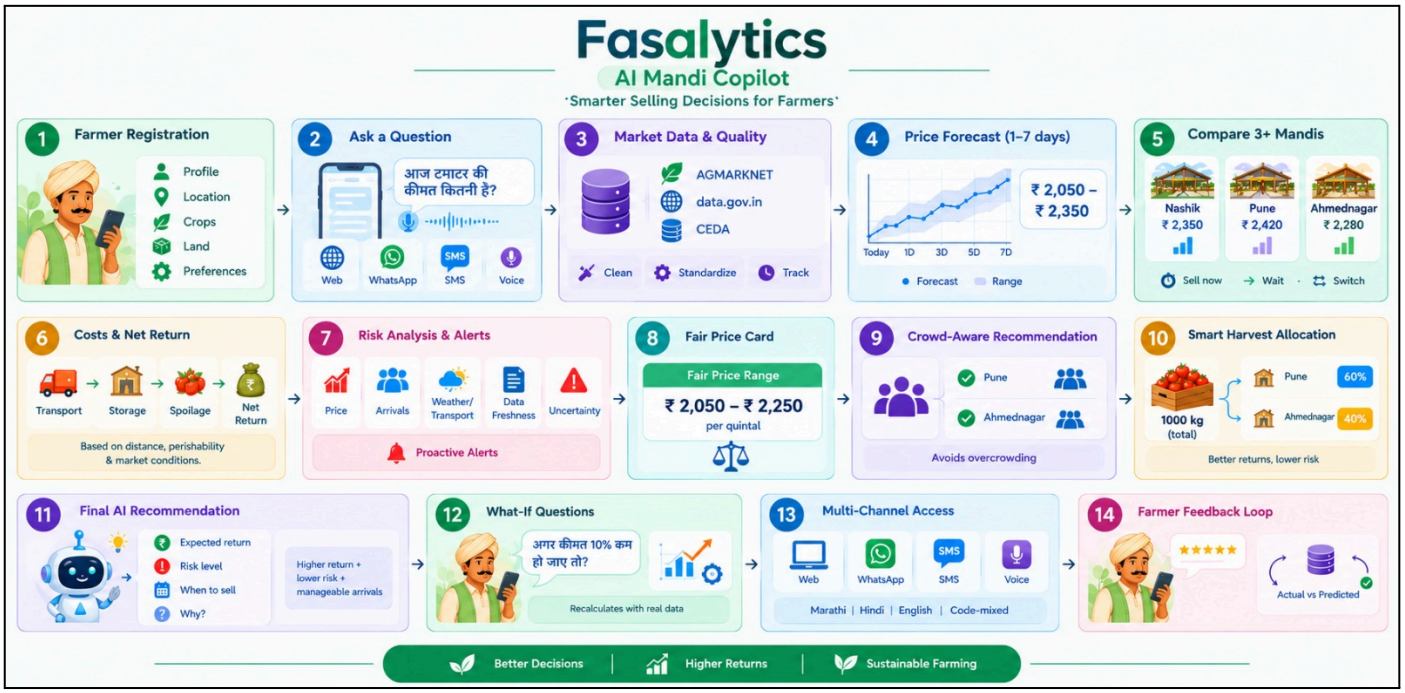
- **Frontend:** React / Next.js, Tailwind CSS, Chart.js
- **Backend:** Node.js, Express.js
- **Authentication:** Firebase
- **Database:** PostgreSQL
- **AI/ML:** Python, Pandas, Scikit-learn, XGBoost, Time-Series Models
- **ML API:** Python + FastAPI
- **LLM & Voice AI:** Gemini / Grok**,** Sarvam AI (Chat, STT, TTS, Translation)
- **Communication:** SMS / WhatsApp APIs
- **External Signals:** Weather APIs

SDG Mapping:

- **SDG 2 – Zero Hunger:** Improves farmer's market access and income.
- **SDG 8 – Decent Work and Economic Growth:** Supports better farmer livelihoods.
- **SDG 12 – Responsible Consumption and Production:** Reduces spoilage and food wastage.







Implementation Screenshots:

